The Alspach–Zhang conjecture: every cubic Cayley graph is 3-edge-colourable

A survey

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Abstract

In 1992 Brian Alspach and Cun-Quan Zhang conjectured that every Cayley graph of valency at least two admits a nowhere-zero 4-flow. By theorems of Mader and Jaeger this is equivalent to the statement that every connected cubic Cayley graph is 3-edge-colourable, or, in the language introduced by Nedela and Škoviera, that there are no *Cayley snarks*. The conjecture is implied by the folklore conjecture that every Cayley graph is hamiltonian, and it is known to hold for solvable groups (Alspach, Liu and Zhang), for all cubic graphs admitting a solvable vertex-transitive group of automorphisms other than the Petersen graph (Potočnik), and for all cubic Cayley graphs on at most 1280 vertices (Potočnik, Spiga and Verret). A minimal counterexample, if one exists, is a Cayley graph on a non-abelian simple group or on a group with a unique non-trivial proper normal subgroup, which has index two (Nedela and Škoviera). We survey these results, give short self-contained proofs of the elementary reductions, record several reformulations (in terms of arc-regular cubic graphs, of bi-Cayley graphs, and of small girth), prove two elementary new tools (a quotient lemma for arbitrary subgroups, with semi-edges, and a description of the 3-edge-colourings of $\mathrm{Cay}(G, \{a, x, x^{-1}\})$ as a-invariant functions $G \to \mathbb{Z}_3$ with a local admissibility condition along the x-cycles), and report a computer verification of the conjecture for all cubic Cayley graphs on 81 non-solvable groups of order up to 259 200, including every group permitted by the Nedela–Škoviera reduction of order less than 262 080; consequently a smallest Cayley snark, if one exists, has more than 262 079 vertices.

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1 Introduction

A *Cayley graph* $\mathrm{Cay}(G, S)$ of a finite group G with respect to a subset $S \subseteq G \setminus \{1\}$ with $S = S^{-1}$ has vertex set G, with g adjacent to gs for every $s \in S$. Cayley graphs are the most symmetric graphs one can build from a group: G itself acts regularly on the vertices by left multiplication. Many of the classical "hard" cubic graphs of graph theory—the Petersen graph, the Coxeter graph, the snarks—are highly symmetric, yet none of them is a Cayley graph. The conjecture surveyed here asserts that this is not an accident.

Conjecture 1.1 (Alspach–Zhang, 1992)**.** *Every connected cubic Cayley graph is* 3*-edge-colourable.*

The conjecture was formulated by Alspach and Zhang at the Louisville workshop on Hamilton cycles in 1992, in the language of flows: *every Cayley graph of valency at least two admits a nowhere-zero* 4*-flow* (see [25, §1], [45, §1]). It first appeared in print in the paper of Alspach, Liu and Zhang [4], who proved it for solvable groups. Since a connected vertex-transitive graph of valency k is k-edge-connected (Mader [27]) and every 4-edge-connected graph has a nowhere-zero 4-flow (Jaeger [18]), the flow version reduces at once to the cubic case, where a nowhere-zero 4-flow is the same thing as a 3-edge-colouring (Tutte [40]); see Proposition 2.5 below. Nedela and Škoviera [33] coined the term *Cayley snark* for a cubic Cayley graph that is not 3-edge-colourable; Conjecture 1.1 says that Cayley snarks do not exist.

The conjecture sits between two much older problems.

- *Snarks and the Four Colour Theorem.* Tait observed in 1880 that the Four Colour Theorem is equivalent to the statement that every bridgeless planar cubic graph is 3-edge-colourable. Non-planar bridgeless cubic graphs need not be: the Petersen graph is the smallest example, and the non-trivial examples are called snarks. A large part of the theory of snarks consists in showing that highly structured cubic graphs are not snarks; the Alspach–Zhang conjecture is the statement of this kind for the strongest possible symmetry assumption compatible with the Petersen graph not being an exception.

- *Hamilton cycles in vertex-transitive graphs.* Lovász asked in 1969 whether every connected vertex-transitive graph has a Hamilton path, and a folklore conjecture asserts that every connected Cayley graph on at least three vertices has a Hamilton cycle [11, 24]. A cubic Cayley graph has even order (Lemma 2.1), so a Hamilton cycle in it is an even cycle whose edges can be coloured alternately, the remaining perfect matching receiving the third colour. Thus *Conjecture 1.1 is a consequence of the hamiltonicity conjecture for cubic Cayley graphs*, and every hamiltonicity result for cubic Cayley graphs is a result about Conjecture 1.1. Only four connected vertex-transitive graphs without a Hamilton cycle are known—the Petersen graph, the Coxeter graph, and their truncations—and none of them is a Cayley graph.

What is known

Table 1 summarises the state of the conjecture. The three main theorems are those of Alspach, Liu and Zhang (solvable groups), Nedela and Škoviera (structure of a minimal counterexample) and Potočnik (solvable vertex-transitive groups); the strongest computational evidence comes from the census of cubic vertex-transitive graphs of Potočnik, Spiga and Verret. We add to this a computation of our own (Section 8) which pushes the verification well beyond the range of the census for the groups singled out by the Nedela–Škoviera reduction: a smallest Cayley snark would have more than 262 079 vertices.

Class of cubic Cayley graphs $\mathrm{Cay}(G, S)$	Status / reference
S consists of three involutions	trivial (colour by generator), §3
$S = \{a, x, x^{-1}\}$ with $\mathrm{ord}(x)$ even	trivial, §3
G abelian, dihedral, nilpotent, supersolvable, of odd-order-times-two, ...	all solvable: [4]
G solvable	Alspach–Liu–Zhang [4]
any cubic graph with a solvable vertex-transitive group of automorphisms, other than the Petersen graph	Potočnik [36]
$\lvert G\rvert \leq 1280$	census [38] (all are hamiltonian)
$G = \langle a, x\rangle$ with $a^2 = x^s = (ax)^3 = 1$ ("$(2, s, 3)$-Cayley graphs")	hamiltonian for s odd [15, 14]; trivial for s even
G has a normal subgroup containing a, or a normal subgroup avoiding a and x whose quotient graph is colourable	Lemmas 2.8, 3.3
minimal counterexample	G non-abelian simple, or G has a unique non-trivial proper normal subgroup T, $\lvert G : T\rvert = 2$, T simple or $T \cong S \times S$ [33]
girth 4; girth 3 or 5 unless $\mathrm{ord}(x) = 3$ or 5	Proposition 3.4
G has a subgroup H avoiding the conjugacy class of x such that the pregraph $H\backslash X$ is 3-edge-colourable	Lemma 4.1
$\mathrm{ord}(x) = 3$ and $\lvert G\rvert \leq 30\,000$	Conder's census [9], Corollary 3.8
every group admissible for a smallest Cayley snark of order $< 262\,080$, and further non-solvable groups of order $\leq 259\,200$	Section 8

Table 1: Known cases of Conjecture 1.1. Throughout, a denotes an involution and x an element of odd order ≥ 3 generating G together with a; by §3 this is the only case that matters.

Organisation

Section 2 fixes terminology and recalls the classical facts about 3-edge-colourings, flows, snarks and coverings that are used throughout. Section 3 disposes of the trivial cases and reduces the conjecture to Cayley graphs $\mathrm{Cay}(G, \{a, x, x^{-1}\})$ with a an involution and x of odd order;

it also contains several elementary reformulations that we have not seen stated explicitly in the literature (arc-regular cubic graphs, bi-Cayley graphs, small girth). Section 4 extends the quotient lemma to arbitrary subgroups (the quotient is then a pregraph with semi-edges) and describes the 3-edge-colourings of $\mathrm{Cay}(G, \{a, x, x^{-1}\})$ as a-invariant functions $G \to \mathbb{Z}_3$ satisfying a local condition along the x-cycles. Section 5 treats solvable groups and Potočnik's generalisation to solvable vertex-transitive groups. Section 6 presents the Nedela–Škoviera reduction with a self-contained proof. Section 7 surveys the hamiltonicity route and the Cayley-map approach of Glover, Kutnar, Malnič, Marušič and Hujdurović. Section 8 describes the computational evidence, including our own. Section 9 collects related problems on flows in Cayley graphs, and Section 10 lists open questions.

2 Preliminaries

2.1 Cayley graphs

Let G be a finite group and $S \subseteq G \setminus \{1\}$ a subset closed under inversion. The Cayley graph $\mathrm{Cay}(G, S)$ has vertex set G and edge set $\{\{g, gs\} : g \in G,\ s \in S\}$. It is a simple $|S|$-regular graph; it is connected if and only if S generates G. Every edge $\{g, gs\}$ carries the label $\{s, s^{-1}\}$, and left multiplication by any $h \in G$ is a label-preserving automorphism, so G acts regularly on the vertices. Conversely, a graph admitting a group of automorphisms acting regularly on its vertices is a Cayley graph on that group. All graphs in this survey are finite; "Cayley graph" will always mean a connected one.

2.2 Cubic Cayley graphs

Let $\mathrm{Cay}(G, S)$ be cubic, so $|S| = 3$. Since $S = S^{-1}$ and $|S|$ is odd, S contains an involution. Either all three elements of S are involutions, or $S = \{a, x, x^{-1}\}$ with $a^2 = 1$ and $\mathrm{ord}(x) \geq 3$. We use the notation

$$X(G; a, x) := \mathrm{Cay}(G, \{a, x, x^{-1}\}), \qquad a^2 = 1, \quad \mathrm{ord}(x) \geq 3, \quad G = \langle a, x \rangle.$$

In $X(G; a, x)$ the a-edges form a perfect matching M_a, and the x-edges form a 2-factor F_x consisting of $|G|/\mathrm{ord}(x)$ cycles of length $\mathrm{ord}(x)$ (the left cosets of $\langle x \rangle$).

Lemma 2.1. *A connected cubic Cayley graph has even order and is either* $\mathrm{Cay}(G, \{a, b, c\})$ *with three involutions a, b, c, or $X(G; a, x)$ as above.*

Proof. The generating set contains an involution, so $|G|$ is even; the dichotomy was explained above. □

2.3 Tait colourings, even 2-factors and flows

A 3-*edge-colouring* (or Tait colouring) of a cubic graph X is a colouring of the edges with three colours such that the three edges at each vertex receive distinct colours; equivalently, a partition of $E(X)$ into three perfect matchings. The following reformulation is used constantly.

Lemma 2.2. *A cubic graph is 3-edge-colourable if and only if it has a 2-factor all of whose cycles are even.*

Proof. Two colour classes of a 3-edge-colouring form a 2-factor of even cycles; conversely, colour the cycles of an even 2-factor alternately and give the complementary perfect matching the third colour. □

In particular a cubic graph with a Hamilton cycle is 3-edge-colourable if and only if its order is even, and a bipartite cubic graph is 3-edge-colourable (Kőnig).

A *nowhere-zero k-flow* on a graph is an orientation together with an assignment of integers from $\{\pm 1, \ldots, \pm(k-1)\}$ to the edges satisfying Kirchhoff's law at every vertex. Tutte [40, 41] showed that a cubic graph admits a nowhere-zero 4-flow if and only if it is 3-edge-colourable, and that a planar graph admits a nowhere-zero 4-flow if and only if its dual is 4-vertex-colourable; his 5-flow conjecture (every bridgeless graph has a nowhere-zero 5-flow) and 3-flow conjecture (every 4-edge-connected graph has a nowhere-zero 3-flow) remain open. Two theorems are needed to relate the flow form of the Alspach–Zhang conjecture to its colouring form.

Theorem 2.3 (Jaeger [18])**.** *Every 4-edge-connected graph admits a nowhere-zero 4-flow.*

Theorem 2.4 (Mader [27])**.** *A connected vertex-transitive graph of valency k is k-edge-connected.*

Proposition 2.5. *The following statements are equivalent.*

(i) *Every connected Cayley graph of valency at least 2 admits a nowhere-zero 4-flow.*

(ii) *Every connected cubic Cayley graph admits a nowhere-zero 4-flow.*

(iii) *Every connected cubic Cayley graph is 3-edge-colourable* (Conjecture 1.1)*.*

Proof. A connected 2-regular graph is a cycle and has a nowhere-zero 2-flow. A Cayley graph of valency $k \geq 4$ is k-edge-connected by Theorem 2.4 and hence has a nowhere-zero 4-flow by Theorem 2.3. So (ii)⇒(i), the converse is trivial, and (ii)⇔(iii) is Tutte's theorem. □

We also recall that a 3-edge-colourable cubic graph has a cycle double cover (take the three 2-factors formed by pairs of colour classes), so Conjecture 1.1 implies the cycle double cover conjecture for cubic Cayley graphs.

2.4 Snarks and cyclic connectivity

A *snark* is usually defined as a cyclically 4-edge-connected cubic graph of girth at least 5 which is not 3-edge-colourable; the connectivity and girth conditions exclude examples obtained trivially from smaller ones. Following [33] we drop these conditions and call *any* cubic Cayley graph that is not 3-edge-colourable a *Cayley snark*. For vertex-transitive cubic graphs the distinction is small:

Theorem 2.6 (Nedela–Škoviera [31])**.** *The cyclic edge-connectivity of a connected cubic vertex-transitive graph equals its girth.*

Thus a Cayley snark of girth at least 5 would be a snark in the classical sense, and Jaeger's conjecture [19, 20] that there are no cyclically 7-edge-connected snarks (still open; the analogous conjecture for girth was refuted by Kochol [21]) would imply Conjecture 1.1 for all cubic Cayley graphs of girth at least 7. We return to this in §3.3.

2.5 Coverings and quotients

A surjective graph homomorphism $\pi\colon \tilde{X} \to X$ between (multi)graphs is a *covering projection* if it maps the set of edges incident with each vertex $\tilde{v}$ bijectively onto the set of edges incident with $\pi(\tilde{v})$. If a group N acts semiregularly on $\tilde{X}$ (every non-identity element acts without fixed vertices and without inverting edges) and the quotient $\tilde{X}/N$ has no loops or semi-edges, then $\tilde{X} \to \tilde{X}/N$ is a covering projection, called a *regular N-covering*.

Lemma 2.7. *If $\pi\colon \tilde{X} \to X$ is a covering projection of cubic multigraphs and X is 3-edge-colourable, then so is $\tilde{X}$.*

Proof. Colour each edge $\tilde{e}$ of $\tilde{X}$ with the colour of $\pi(\tilde{e})$. At each vertex $\tilde{v}$ the three edges map bijectively onto the three (distinctly coloured) edges at $\pi(\tilde{v})$. $\square$

For Cayley graphs the relevant quotients come from normal subgroups. Let $N \trianglelefteq G$ and write $\bar{g} = gN$.

Lemma 2.8 (quotient lemma)**.** *Let $X = X(G; a, x)$ with $\operatorname{ord}(x)$ odd and let $1 \neq N \trianglelefteq G$ be a proper normal subgroup with $a \notin N$ and $x \notin N$. Then $\bar{G} = G/N$ is generated by the involution $\bar{a}$ and the element $\bar{x}$ of odd order at least 3, $\bar{X} = X(\bar{G}; \bar{a}, \bar{x})$ is a connected cubic Cayley graph on a group of order $|G|/|N|$, and $g \mapsto \bar{g}$ is a regular N-covering $X \to \bar{X}$. Consequently, if $\bar{X}$ is 3-edge-colourable then so is X.*

Proof. $\bar{a} \neq 1$ since $a \notin N$, and $\operatorname{ord}(\bar{x})$ divides the odd number $\operatorname{ord}(x)$ and is > 1 since $x \notin N$; hence $\operatorname{ord}(\bar{x}) \geq 3$, $\bar{x} \neq \bar{x}^{-1}$ and $\bar{x} \neq \bar{a}$. So $\{\bar{a}, \bar{x}, \bar{x}^{-1}\}$ is an inverse-closed 3-subset of $\bar{G} \setminus \{1\}$ generating $\bar{G}$. The map $g \mapsto \bar{g}$ sends the three edges $\{g, ga\}, \{g, gx\}, \{g, gx^{-1}\}$ at g to the three distinct edges $\{\bar{g}, \bar{g}\bar{a}\}, \{\bar{g}, \bar{g}\bar{x}\}, \{\bar{g}, \bar{g}\bar{x}^{-1}\}$ at $\bar{g}$, and N acts semiregularly by left multiplication with quotient $\bar{X}$. Apply Lemma 2.7. $\square$

When N contains a or x the quotient degenerates (the a-edges become semi-edges, or the x-edges become loops). These two cases behave very differently: the first is harmless (Lemma 3.3), the second is the heart of the problem (§3.5).

3 The easy cases and some reformulations

3.1 Trivial colourings

Proposition 3.1. *Let X be a connected cubic Cayley graph.*

(a) *If $X = \operatorname{Cay}(G, \{a, b, c\})$ with a, b, c involutions, then colouring each edge by its generator is a 3-edge-colouring.*

(b) *If $X = X(G; a, x)$ with $\operatorname{ord}(x)$ even, then F_x is an even 2-factor and X is 3-edge-colourable.*

(c) *If G is abelian, then X is 3-edge-colourable.*

Proof. (a) and (b) are immediate from Lemma 2.2. For (c), by (a) and (b) we may assume $X = X(G; a, x)$ with $\operatorname{ord}(x)$ odd; as G is abelian, $G = \langle x \rangle \times \langle a \rangle$ and X is the prism $C_{\operatorname{ord}(x)} \square K_2$, whose a-edges together with the "rungs" form a Hamilton cycle of even length. $\square$

Corollary 3.2. *Conjecture 1.1 is equivalent to the following statement: for every finite group $G = \langle a, x \rangle$ generated by an involution a and an element x of odd order $s \geq 3$, the graph $X(G; a, x)$ is 3-edge-colourable.*

From now on we therefore assume that $X = X(G; a, x)$ with $s = \operatorname{ord}(x)$ odd, $s \geq 3$. Note that X then contains the odd cycles of F_x, so X is never bipartite, and a 3-edge-colouring must use a 2-factor different from the "obvious" one. In terms of the group, X is 3-edge-colourable if and only if there is a subset $M \subseteq G$ (the vertices whose a-edge is *not* used by the even 2-factor) such that the 2-factor consisting of the a-edges at $G \setminus M$ and suitable x-edges at M has only even cycles; there seems to be no useful algebraic characterisation of this condition.

3.2 Normal subgroups containing the involution

Lemma 3.3. *Let $X = X(G; a, x)$ with $\operatorname{ord}(x)$ odd. If G has a proper normal subgroup N with $a \in N$ (equivalently, if there is a surjective homomorphism $\varphi\colon G \to \mathbb{Z}_m$, $m \geq 3$, with $\varphi(a) = 0$), then X is 3-edge-colourable.*

Proof. $G/N = \langle xN \rangle$ is cyclic of order $m \mid \operatorname{ord}(x)$, and $m \geq 3$ because $N \neq G$; let $\varphi\colon G \to \mathbb{Z}_m$ be the quotient map, so $\varphi(a) = 0$ and $\varphi(x) = 1$. Choose a proper vertex 3-colouring $c\colon \mathbb{Z}_m \to \{1, 2, 3\}$ of the cycle C_m (it exists for every $m \geq 3$). Colour the x-edge $\{g, gx\}$ with $c(\varphi(g))$ and the a-edge $\{g, ga\}$ with the colour in $\{1, 2, 3\} \setminus \{c(\varphi(g) - 1), c(\varphi(g))\}$; the latter is well defined because $\varphi(ga) = \varphi(g)$. At a vertex g with $\varphi(g) = i$ the two x-edges $\{gx^{-1}, g\}$ and $\{g, gx\}$ have the distinct colours $c(i - 1)$ and $c(i)$, and the a-edge has the third colour. □

Note that m is automatically odd. The lemma says that the "degenerate quotient" in which the a-edges collapse to semi-edges is never an obstacle.

3.3 Small girth

A cycle of length ℓ through the identity of $X = X(G; a, x)$ corresponds to a cyclically reduced word of length ℓ in a, x, x^{-1} (no factor aa, xx^{-1} or $x^{-1}x$, also cyclically) representing 1 in G.

Proposition 3.4. *Let $X = X(G; a, x)$ with $s = \operatorname{ord}(x)$ odd, $s \geq 3$, and G not cyclic.*

(a) *X has girth 3 if and only if $s = 3$.*

(b) *If X has girth 4, then $\langle x \rangle \trianglelefteq G$ and $G \cong D_{2s}$ is dihedral.*

(c) *If X has girth 5, then either $s = 5$ or $\langle x \rangle \trianglelefteq G$, so that $G = \langle x \rangle \rtimes \langle a \rangle$ is metacyclic.*

In particular Conjecture 1.1 holds for all cubic Cayley graphs of girth 4, and for those of girth 3 (resp. 5) unless $s = 3$ (resp. $s = 5$).

Proof. A cyclically reduced word in $a, x^{\pm 1}$ of length $\ell \leq 5$ has one of the forms $x^{\pm \ell}$; ax^k with $|k| = \ell - 1$; or (for $\ell \in \{4, 5\}$) $ax^i ax^j$ with $i, j \neq 0$ and $|i| + |j| = \ell - 2$. A relation $x^{\pm \ell} = 1$ with $\ell \leq 5$ and s odd forces $s = \ell \in \{3, 5\}$. A relation $ax^k = 1$ gives $a \in \langle x \rangle$ and G cyclic, which is excluded. A relation $ax^i ax^j = 1$ gives $ax^i a = x^{-j}$ with $|i|, |j| \in \{1, 2, 3\}$ and $|i| + |j| \leq 3$; since s is odd, x^i and x^{-j} generate $\langle x \rangle$ unless $s = 3$ and $3 \mid i$ or $3 \mid j$, which would give girth 3. Hence a normalises $\langle x \rangle$, $\langle x \rangle \trianglelefteq G$, and $G = \langle x \rangle \rtimes \langle a \rangle$. For $\ell = 4$ the only such relations are $(ax)^2 = 1$ (giving D_{2s}) and $axax^{-1} = 1$ (giving G abelian, hence cyclic). The final assertion follows from Theorem 5.1 below, since metacyclic groups are solvable. □

Combined with Theorem 2.6, this shows that in a proof of Conjecture 1.1 by "girth" the cases requiring an idea are girth 3 with $s = 3$, girth 5 with $s = 5$, girth 6, and girth at least 7; a proof of Jaeger's conjecture on cyclically 7-edge-connected snarks would settle the last case. This parallels the observation of Aboomahigir and Nedela [1] that, modulo Thomassen's conjecture that cyclically 7-connected cubic graphs other than the Coxeter graph are hamiltonian, the hamiltonicity of cubic Cayley graphs reduces to girth at most 6.

3.4 Generators of order three: arc-regular cubic graphs

The case $s = 3$ has a pleasant reformulation. Recall that the *truncation* $\mathrm{T}(Y)$ of a cubic graph Y is obtained by replacing every vertex by a triangle; its vertices are the arcs (ordered pairs of adjacent vertices) of Y, two arcs being adjacent if they have the same initial vertex or are inverse to each other.

Lemma 3.5. *A cubic multigraph Y is 3-edge-colourable if and only if $\mathrm{T}(Y)$ is.*

Proof. In a 3-edge-colouring of $\mathrm{T}(Y)$ the three edges of a triangle receive three distinct colours, hence so do the three edges leaving the triangle; restricting the colouring to the edges of $\mathrm{T}(Y)$ that come from edges of Y gives a 3-edge-colouring of Y. Conversely, given a 3-edge-colouring of Y, colour each triangle edge of $\mathrm{T}(Y)$ with the colour of the edge of Y opposite to it in the triangle. $\square$

A group $G \le \mathrm{Aut}(Y)$ is *arc-regular* if it acts regularly (transitively with trivial stabilisers) on the arcs of Y.

Proposition 3.6. *Let $G = \langle a, y\rangle$ with $a^2 = y^3 = 1$ and $|G| > 6$. Let Y be the coset graph with vertex set $\{\langle y\rangle g : g \in G\}$ in which $\langle y\rangle g$ is adjacent to $\langle y\rangle a y^i g$ for $i = 0, 1, 2$. Then Y is a connected simple cubic graph on $|G|/3$ vertices, G acts on it by right multiplication as an arc-regular group of automorphisms, and $X(G; a, y) \cong \mathrm{T}(Y)$. Conversely, if Y is a connected cubic graph with an arc-regular group G of automorphisms, then $G = \langle a, y\rangle$ for an involution a and an element y of order 3, and $\mathrm{T}(Y) \cong X(G; a, y)$.*

Proof. Put $H = \langle y\rangle$. The three neighbours Ha, Hay, Hay^2 of the vertex H are distinct: $Ha = Hay^i$ would give $ay^ia^{-1} \in H$, so that a normalises H and $|G| \le 6$. Right multiplication by G preserves adjacency and is transitive on vertices; the stabiliser H of the vertex H permutes its three neighbours cyclically ($Hay^i \cdot y = Hay^{i+1}$), so G is transitive on arcs, and $|G| = 3\,|V(Y)|$ equals the number of arcs, so the action on arcs is regular. Identify the arc (Hg, Hag) with $g \in G$; arcs with the same initial vertex correspond to g and $y^{\pm 1}g$, and the inverse of the arc g is the arc ag. So $\mathrm{T}(Y)$ is the Cayley graph of G with respect to $\{a, y, y^{-1}\}$ and *left* multiplication, which is isomorphic to $X(G; a, y)$ via $g \mapsto g^{-1}$.

Conversely, if G is arc-regular on Y, the stabiliser G_v of a vertex v acts regularly on the three arcs leaving v, so $G_v = \langle y\rangle \cong \mathbb{Z}_3$; some $a \in G$ maps the arc (v, w) to (w, v), and a^2 fixes (v, w), so a is an involution. Connectedness of Y gives $G = \langle G_v, a\rangle$, and identifying arcs with group elements as above gives $\mathrm{T}(Y) \cong X(G; a, y)$. $\square$

Corollary 3.7. *Conjecture 1.1 for generating pairs (a, x) with $\mathrm{ord}(x) = 3$ is equivalent to the statement that every connected cubic graph admitting an arc-regular group of automorphisms is 3-edge-colourable.*

Conder's census [9] of all connected cubic arc-transitive graphs on at most 10 000 vertices records that every one of them other than the Petersen graph and the Coxeter graph has a Hamilton cycle. Since the Coxeter graph is 3-edge-colourable, Lemma 3.5 and Proposition 3.6 give:

Corollary 3.8. *Every connected cubic graph on at most 10 000 vertices that admits an arc-regular group of automorphisms is 3-edge-colourable. Equivalently, Conjecture 1.1 holds for every $X(G; a, x)$ with $\mathrm{ord}(x) = 3$ and $|G| \le 30\,000$.*

The Petersen graph is arc-transitive with automorphism group S_5, but has no arc-regular subgroup (S_5 has no subgroup of order 30), in accordance with the conjecture; the same applies to its truncation, which is a vertex-transitive cubic graph that is not 3-edge-colourable but not a Cayley graph. Cubic arc-transitive graphs have been classified into seventeen "action types" by Conder and Nedela [10] according to which of the Djoković–Miller groups [12] occur as subgroups of the automorphism group; the ones containing an arc-regular ("1-regular") subgroup are exactly the graphs in Corollary 3.7. Since every finite non-abelian simple group, apart from the symplectic groups $\mathrm{PSp}(4, 2^f)$, $\mathrm{PSp}(4, 3^f)$ and finitely many others (among them A_6, A_7, A_8, $\mathrm{PSL}(3, 4)$, $\mathrm{PSU}(3, 3)$, $\mathrm{PSU}(3, 5)$, $\mathrm{PSU}(4, 3)$, $\mathrm{PSU}(5, 2)$ and the sporadic groups M_{11}, M_{22}, M_{23}, McL), is generated by an involution and an element of order 3 (Liebeck and Shalev [26], Woldar [42]; see [35] for the classical groups of small dimension), the case $s = 3$ alone already concerns a cubic Cayley graph on almost every simple group. (In our computations,

Table 2, the groups A_6, A_7, A_8, PSL(3, 4), PSU(3, 3) and M_{11} indeed have no generating pair with $\mathrm{ord}(x) = 3$.)

3.5 Normal subgroups of index two: bi-Cayley graphs

Suppose that $X = X(G; a, x)$ with $s = \mathrm{ord}(x)$ odd and that $T \trianglelefteq G$ has index 2. Then $x \in T$ (its image in $G/T \cong \mathbb{Z}_2$ has odd order) and $a \notin T$, so $G = T \rtimes \langle a \rangle$ and conjugation by a induces an involutory automorphism α of T with $T = \langle x, \alpha(x) \rangle$. The vertex set of X is $T \sqcup Ta$, the x-edges inside T are the cycles of x, the x-edges inside Ta are, after the identification $ta \leftrightarrow t$, the cycles of $\alpha(x)$ (because $ta \cdot x = t\,\alpha(x)\,a$), and the a-edges form the identity matching $t \leftrightarrow ta$. Thus

$$X \;\cong\; B(T; x, \alpha(x)), \tag{1}$$

where, for a group T and elements $u, v \in T$, the graph $B(T; u, v)$ has vertex set $T \times \{0, 1\}$ and edges $(t, 0) \sim (tu, 0)$, $(t, 1) \sim (tv, 1)$ and $(t, 0) \sim (t, 1)$ for all $t \in T$. This is a *bi-Cayley graph* of T, a natural generalisation of the generalised Petersen graphs $GP(n, k) = B(\mathbb{Z}_n; 1, k)$. Conversely, for any group T, any $x \in T$ of odd order and any involutory automorphism α of T with $\langle x, \alpha(x) \rangle = T$, the graph $B(T; x, \alpha(x))$ is the cubic Cayley graph $X(T \rtimes \langle \alpha \rangle; \alpha, x)$.

The generalised Petersen graph $GP(n, k)$ is a Cayley graph if and only if $k^2 \equiv 1 \pmod{n}$ (Nedela and Škoviera [32]), which is exactly the condition that multiplication by k be an involutory automorphism of $\mathbb{Z}_n$; and Castagna and Prins [5] proved Watkins' conjecture that every generalised Petersen graph other than the Petersen graph itself is 3-edge-colourable. (The Petersen graph is $GP(5, 2)$ with $2^2 \equiv -1$, not $+1$, modulo 5.) So Conjecture 1.1 holds for all generalised Petersen graphs, and the results of §6 show that the whole conjecture reduces to graphs of the form (1) with T simple or $T \cong S \times S$, together with Cayley graphs on simple groups.

4 Quotients by arbitrary subgroups, and a reformulation over $\mathbb{Z}_3$

This section contains two elementary observations which we have not found in the literature in this form. The first extends the quotient lemma (Lemma 2.8) from normal subgroups to arbitrary subgroups, at the price of allowing semi-edges in the quotient; it is the tool behind the computations of Section 8. The second describes the 3-edge-colourings of $X(G; a, x)$ as the a-invariant functions $G \to \mathbb{Z}_3$ satisfying a local condition along the x-cycles.

4.1 Pregraph quotients

A *pregraph* is a graph in which, besides ordinary edges, *semi-edges* (edges with only one end) and loops are allowed; each vertex of a cubic pregraph is incident with three *darts*, an ordinary edge contributing a dart at each of its ends and a semi-edge one dart. A 3-edge-colouring of a pregraph colours the ordinary edges and the semi-edges so that the three darts at every vertex carry three different colours.

Let $X = X(G; a, x)$ with $s = \mathrm{ord}(x)$ odd and let $H \leq G$ be *any* subgroup. The left action of H on $V(X) = G$ is free, and so is its action on the darts (g, gs), $s \in \{a, x, x^{-1}\}$. The quotient pregraph $H\backslash X$ has vertex set $H\backslash G = \{Hg\}$ and one dart for each H-orbit of darts; the darts (g, gs) and (gs, g) lie in the same orbit if and only if $hg = gs$ and $hgs = g$ for some $h \in H$, i.e. $s^2 = 1$ and $gsg^{-1} \in H$, which happens only for $s = a$. Hence the a-edge at Hg is a semi-edge exactly when $gag^{-1} \in H$, and the x-edge at Hg is a loop exactly when $gxg^{-1} \in H$.

Lemma 4.1 (generalised quotient lemma)**.** *Let $X = X(G; a, x)$ with $\mathrm{ord}(x)$ odd and let $H \leq G$ be a subgroup containing no conjugate of x. Then $\bar{X} = H\backslash X$ is a loopless cubic pregraph on*

$|G:H|$ vertices with exactly $|H \cap a^G| \cdot |C_G(a)|/|H|$ semi-edges, and $X \to \bar{X}$, $g \mapsto Hg$, is a covering of pregraphs. If $\bar{X}$ is 3-edge-colourable, then so is X; more precisely, the 3-edge-colourings of $\bar{X}$ correspond bijectively to the 3-edge-colourings of X that are invariant under left multiplication by H.

Proof. The three darts at g are mapped to the three darts at Hg, which are distinct because $\bar{X}$ has no loops; so the projection is a covering. Given a colouring of $\bar{X}$, colour every dart of X with the colour of its image; the two darts of an ordinary edge $\{g, ga\}$ with $gag^{-1} \notin H$ map to the two darts of an ordinary edge of $\bar{X}$ and receive the same colour, and if $gag^{-1} \in H$ they map to the same semi-edge. Thus the colouring is a well defined 3-edge-colouring of X, obviously H-invariant, and conversely an H-invariant colouring of X is constant on H-orbits of darts and descends to $\bar{X}$. The count of semi-edges is the number of cosets Hg with $gag^{-1} \in H$, i.e. the number of elements g with $gag^{-1} \in H \cap a^G$, which is $|H \cap a^G|\,|C_G(a)|$, divided by $|H|$. □

Lemma 2.8 is the case $H \trianglelefteq G$ with $a \notin H$, and Lemma 2.7 in the form used by the program of §8 is the case of odd $|H|$ (no semi-edges). The point of Lemma 4.1 is that H may now be large: any subgroup whose order is not divisible by $\mathrm{ord}(x)$, the centraliser $C_G(a)$ whenever no conjugate of x centralises a (the quotient is then the pregraph on the conjugacy class a^G in which b is joined to b^a, b^x and $b^{x^{-1}}$), a point stabiliser of any permutation representation in which x has no fixed points, and so on. Semi-edges carry a parity constraint which, in practice, is the main reason why the largest admissible subgroup need not work.

Lemma 4.2 (parity)**.** *Let $\bar{X}$ be a cubic pregraph with n vertices and σ semi-edges. If $\bar{X}$ is 3-edge-colourable then the semi-edges fall into three colour classes each of size $\equiv n \pmod 2$; in particular $\sigma \equiv n \pmod 2$, and $\sigma \geq 3$ if n is odd.*

Proof. Each colour class contains e ordinary edges and σ' semi-edges with $2e + \sigma' = n$. □

Example 4.3. Let $G = \mathrm{PSL}(2, q)$ act on the projective line $\Omega = \mathrm{GF}(q) \cup \{\infty\}$, and let x lie in a non-split torus, so that x has no fixed point on Ω and $H = G_\infty$ (a Borel subgroup) is admissible in Lemma 4.1; the quotient is the Schreier graph of $\langle a, x\rangle$ on the $q+1$ points. If q is even, an involution a fixes exactly one point, so $\bar{X}$ has $q+1$ (odd) vertices and one semi-edge, and Lemma 4.2 shows that $\bar{X}$ is never 3-edge-colourable—although X is, for $q \in \{4, 8, 16, 32\}$ (Table 2). If $q \equiv 3 \pmod 4$ the involutions are fixed-point-free on Ω and if $q \equiv 1 \pmod 4$ they have two fixed points, so the parity condition holds; for these q we found by computer that the $(q+1)$-vertex quotient is 3-edge-colourable for most but not all generating pairs (for instance for all 192 generating pairs with $\mathrm{ord}(x) = 13$ in $\mathrm{PSL}(2, 25)$ or $\mathrm{ord}(x) = 15$ in $\mathrm{PSL}(2, 29)$, but for only 11 of the 13 generating pairs with $\mathrm{ord}(x) = \mathrm{ord}(ax) = 7$ in $\mathrm{PSL}(2, 13)$).

4.2 Hamiltonian quotients

The quotient $H\backslash X$ is itself "hamiltonian" when $\langle x\rangle$ acts transitively on $H\backslash G$, i.e. when $G = H\langle x\rangle$ is an exact factorisation; the x-cycle through H then passes through all $|G:H| = \mathrm{ord}(x)$ vertices. For such pregraphs there is a simple sufficient condition.

Lemma 4.4. *Let Y be a loopless cubic pregraph with a Hamilton cycle C. If $|C|$ is even, or if two adjacent vertices of C carry semi-edges, then Y is 3-edge-colourable.*

Proof. If $|C|$ is even, colour the edges of C alternately $0, 1$ and everything else 2. Otherwise let u, u' be adjacent vertices of C with semi-edges; colour the edge uu' with 2, the path $C - uu'$ alternately $0, 1$, every chord with 2 and the semi-edges at u, u' with the colours missing there. Every vertex other than u, u' sees the colours $0, 1$ on C, so this is proper. □

Proposition 4.5. *Let $X = X(G; a, x)$ with $\mathrm{ord}(x)$ odd. Suppose $G = H\langle x\rangle$ with $H \cap \langle x\rangle = 1$ for a subgroup H containing a and xax^{-1}. Then X is 3-edge-colourable.*

Proof. Since $\langle x \rangle$ acts regularly on $H\backslash G$, x has no fixed point there, so H contains no conjugate of x and Lemma 4.1 applies; the pregraph $H\backslash X$ has the Hamilton cycle $H, Hx, Hx^2, \ldots$, and its adjacent vertices H and Hx carry semi-edges because $a \in H$ and $xax^{-1} \in H$. Apply Lemma 4.4. □

For instance, if n is odd, x is an n-cycle in S_n and a is an involution fixing two cyclically consecutive points of x (say $a = (1\,2)$, or any involution with at least $(n+1)/2$ fixed points), then Proposition 4.5 with $H = S_{n-1}$ shows that $X(\langle a, x \rangle; a, x)$ is 3-edge-colourable; for $a = (1\,2)$ this is a two-line proof of the colouring consequence of the Compton–Williamson theorem quoted in §7. The adjacency hypothesis cannot simply be replaced by "at least three semi-edges" (the minimum allowed by Lemma 4.2): the Petersen graph minus a vertex is a 9-cycle with three chords and three vertices of valency two, i.e. the pregraph $S_8\backslash X$ for $x = (0\,1\,\cdots\,8)$ and $a = (1\,5)(2\,7)(4\,8)$, and it is not 3-edge-colourable, although $\langle a, x \rangle$ (a solvable group of order 162) and X are perfectly harmless. This example illustrates the main difficulty of the quotient method: a small "natural" quotient of a colourable Cayley graph can be a Petersen-like snark, so a proof of Conjecture 1.1 along these lines must be able to choose the subgroup H adaptively.

4.3 The colourings as functions $G \to \mathbb{Z}_3$

Let $X = X(G; a, x)$ with $s = \mathrm{ord}(x)$ odd, and identify the three colours with the elements of $\mathbb{Z}_3$, so that the three colours at a vertex sum to 0. Given a 3-edge-colouring κ of X, let $t(g) \in \mathbb{Z}_3$ be the colour of the a-edge at g, i.e. the colour missing from the two x-edges at g. Then $t(ga) = t(g)$, and κ is determined by t: on the x-cycle through g, the colour of the edge $\{gx^i, gx^{i+1}\}$ is $-\big(t(gx^i) + \kappa(\{gx^{i-1}, gx^i\})\big)$, and the following lemma shows that t determines κ on the cycle without any initial choice. We say that a function $t\colon \mathbb{Z}_s \to \mathbb{Z}_3$ is *admissible* if it is the sequence of colours of the pendant edges in some proper 3-edge-colouring of the cycle C_s (with a pendant edge attached at each vertex), equivalently the sequence of colours missing at the vertices of C_s.

Lemma 4.6. *Let $s \geq 3$ be odd and $t\colon \mathbb{Z}_s \to \mathbb{Z}_3$. Call $i \in \mathbb{Z}_s$ a* change *if $t(i+1) \neq t(i)$, and put $\delta_i = t(i+1) - t(i) \in \{\pm 1\}$ for a change i. Then t is admissible if and only if t is not constant and, for any two cyclically consecutive changes i and j (so that t is constant on $i+1, \ldots, j$), $\delta_i = \delta_j$ holds precisely when $j - i$ is odd. Each admissible t comes from exactly one 3-edge-colouring of C_s; in particular there are $2^s - 2$ admissible functions.*

Proof. Let $f_i \in \mathbb{Z}_3$ be the colour of the edge $\{i, i+1\}$ of C_s in a proper 3-edge-colouring, so $t(i) = -(f_{i-1} + f_i)$, and put $\epsilon_i = f_i - f_{i-1} \in \{\pm 1\}$. Then $t(i+1) - t(i) = -(\epsilon_i + \epsilon_{i+1})$, so i is a change if and only if $\epsilon_i = \epsilon_{i+1}$, in which case $\delta_i = -2\epsilon_i = \epsilon_i$, while $\epsilon_{i+1} = -\epsilon_i$ at a non-change. Between consecutive changes $i < j$ the signs $\epsilon_{i+1}, \ldots, \epsilon_j$ alternate, so $\delta_j = \epsilon_j = (-1)^{j-i-1}\epsilon_{i+1} = (-1)^{j-i-1}\delta_i$, which is the stated rule; and t is not constant since s is odd. Conversely, let t be non-constant and satisfy the rule. Define $\epsilon_i = \epsilon_{i+1} = \delta_i$ at every change i and let ϵ alternate between changes; the rule says exactly that this is consistent around the cycle. Put $f_i = f_0 + \epsilon_1 + \cdots + \epsilon_i$; this is a proper colouring of C_s provided $\epsilon_1 + \cdots + \epsilon_s \equiv 0 \pmod 3$, and indeed $\sum_i \epsilon_i \equiv -2\sum_i \epsilon_i = \sum_i \big(t(i+1) - t(i)\big) = 0$. By construction $-(f_{i-1} + f_i)$ and $t(i)$ have the same differences, so they differ by a constant c, and replacing f by $f + c$ (which changes $-(f_{i-1} + f_i)$ by $-2c = c$) makes them equal. Since ϵ and c are forced, the colouring is unique; the count follows because C_s has $2^s - 2$ proper 3-edge-colourings for odd s. □

Proposition 4.7. *Let $X = X(G; a, x)$ with $s = \mathrm{ord}(x)$ odd. The 3-edge-colourings of X (with colours in $\mathbb{Z}_3$) are in bijection with the functions $t\colon G \to \mathbb{Z}_3$ such that $t(ga) = t(g)$ for all g and, for every g, the sequence $i \mapsto t(gx^i)$ on $\mathbb{Z}_s$ is admissible.*

Proof. The map $\kappa \mapsto t$ described above is injective by Lemma 4.6, and conversely, given t, colouring each x-cycle by the unique colouring of Lemma 4.6 and each a-edge $\{g, ga\}$ with $t(g) = t(ga)$ gives a proper colouring: at g the two x-edges have the colours other than $t(g)$. □

Thus a 3-edge-colouring of X is the same thing as a 3-colouring of the perfect matching M_a (of the $|G|/2$ a-edges) such that, along every x-cycle, the colours of the a-edges met form an admissible sequence: a constraint satisfaction problem with $|G|/2$ variables and $|G|/s$ constraints of arity s, each variable occurring in two constraints. The simplest admissible sequences are those with exactly three changes, all with the same δ: t takes the values $c, c+1, c+2$ on three consecutive arcs of odd lengths. Lemma 3.3 is the case in which t is pulled back from a valid function on the quotient cycle $\mathbb{Z}_m = G/N$. Two small cases deserve to be spelled out.

Corollary 4.8. *Let $X = X(G; a, x)$.*

(a) *If* $\mathrm{ord}(x) = 3$*, then X is* 3*-edge-colourable if and only if there is $t\colon G \to \mathbb{Z}_3$ with $t(ga) = t(g)$ which takes three different values on every triangle $\{g, gx, gx^2\}$, i.e. if and only if the* 4*-valent graph X/M_a obtained by contracting the a-edges has a proper* 3*-vertex-colouring. (In the notation of §3.4, X/M_a is the line graph of the arc-regular graph Y, so this is Lemma 3.5 again.)*

(b) *If* $\mathrm{ord}(x) = 5$*, then X is* 3*-edge-colourable if and only if there is $t\colon G \to \mathbb{Z}_3$ with $t(ga) = t(g)$ such that on every x-cycle t is constant on three consecutive vertices and takes the two other values on the remaining two vertices.*

Proof. For $s = 3$ a sequence is admissible if and only if its three values are distinct (there are $6 = 2^3 - 2$ of them). For $s = 5$ the $30 = 2^5 - 2$ admissible sequences are exactly the rotations of (c, c, c, c', c'') with $\{c, c', c''\} = \mathbb{Z}_3$: these are admissible by Lemma 4.6 (three changes at mutual distances $3, 1, 1$, all with the same δ), and there are $5 \cdot 6 = 30$ of them. □

Proposition 4.7 combines with Lemma 4.1: an H-invariant colouring is a function t on $H\backslash G$, constant on the orbits of a, admissible on the cycles of x. For the coset space of a large subgroup this is a very small constraint satisfaction problem, and it is the form in which the SAT solver of Section 8 sees the problem. We have not been able to turn either formulation into a proof of Conjecture 1.1 for any infinite family of simple groups; the difficulty is visible already in Example 4.3.

5 Solvable groups

Theorem 5.1 (Alspach, Liu, Zhang [4])**.** *Every connected cubic Cayley graph on a solvable group is* 3*-edge-colourable. Equivalently, every connected Cayley graph of valency at least* 2 *on a solvable group admits a nowhere-zero* 4*-flow.*

The proof is an induction along a normal series. With the lemmas of §§2–3 in hand it can be organised as follows. Let $X = X(G; a, x)$ with G solvable and $\mathrm{ord}(x)$ odd, and let N be a minimal normal subgroup of G; it is elementary abelian. If $a \notin N$ and $x \notin N$, Lemma 2.8 reduces to the smaller solvable group G/N. If $a \in N$, Lemma 3.3 applies (note $N \neq G$ as G is not abelian of prime order). Otherwise $x \in N$, N has index 2 and $G = N \rtimes \langle a \rangle$ with N an elementary abelian p-group (p odd) generated by x and x^a; thus $N \cong \mathbb{Z}_p$ or $\mathbb{Z}_p^2$, and X is either the prism over C_p (if $x^a = x^{\pm 1}$), or the bi-Cayley graph $B(\mathbb{Z}_p^2; (1,0), (0,1))$; the prism is hamiltonian, and the second graph is 3-edge-colourable as well (see [4] for the details in the original formulation; we also confirmed this by computer for $p \leq 13$).

Theorem 5.1 covers, among others, all cubic Cayley graphs on abelian, nilpotent, supersolvable, dihedral and generalised dihedral groups, on p-groups, on groups of order $p^a q^b$ (Burnside) and on all groups of order less than 60.

5.1 Solvable vertex-transitive groups

Potočnik extended Theorem 5.1 from regular to transitive actions of solvable groups.

Theorem 5.2 (Potočnik [36])**.** *Let X be a connected cubic simple graph whose automorphism group contains a solvable subgroup acting transitively on the vertices. If X is not 3-edge-colourable, then X is the Petersen graph.*

The Petersen graph is a genuine exception: the subgroup of S_5 of order 20 (the affine group $\mathrm{AGL}(1,5)$) is solvable and acts transitively on its ten vertices. Theorem 5.2 contains Theorem 5.1, since a Cayley graph on a solvable group admits that group as a solvable regular, hence transitive, group of automorphisms, and the Petersen graph is not a Cayley graph.

The strategy of the proof is again inductive, using the theory of elementary abelian regular covers developed by Malnič, Marušič and Potočnik [28, 29]. If $G \leq \mathrm{Aut}(X)$ is solvable and vertex-transitive, a minimal normal subgroup N of G is elementary abelian; if N acts semiregularly on $V(X)$ then X is a regular N-cover of the quotient X/N, which admits the solvable vertex-transitive group G/N, and Lemma 2.7 applies provided X/N is a cubic graph. The remaining cases (non-semiregular N, quotients of smaller valency) are analysed directly, and the base of the induction involves the Petersen graph: as Malnič and Potočnik put it, "the fact that all elementary abelian covers of the Petersen graph are 3-edge-colourable was used ... in order to prove, by induction, that every connected cubic graph admitting a vertex-transitive solvable group of automorphisms (with the sole exception of the Petersen graph itself) is 3-edge-colourable" [29, §1].

Theorem 5.2 also yields the conjecture for many non-Cayley vertex-transitive cubic graphs, and for every cubic vertex-transitive graph whose order n is such that all transitive permutation groups of degree n are solvable.

6 The structure of a minimal counterexample

Nedela and Škoviera proved that a smallest Cayley snark, if one exists, must come from a group that is "almost as simple as possible". A group G is *almost simple* if $T \leq G \leq \mathrm{Aut}(T)$ for some non-abelian simple group T.

Theorem 6.1 (Nedela–Škoviera [33])**.** *Let $X = X(G; a, x)$ be a Cayley snark of smallest possible order. Then either*

(i) *G is a non-abelian simple group; or*

(ii) *G has exactly one non-trivial proper normal subgroup T; moreover $|G : T| = 2$, $x \in T$, $a \notin T$, and T is either a non-abelian simple group (so that G is almost simple) or the direct product $S \times S$ of two isomorphic non-abelian simple groups, the involution a interchanging the two factors (so that $G \cong S \wr \mathbb{Z}_2$).*

In particular every Cayley snark is a regular cover of a Cayley snark on a group of type (i) *or* (ii)*.*

The last sentence follows from the first by repeated application of Lemma 2.8 to a Cayley snark that is not minimal (the quotient by a normal subgroup avoiding a and x is again a Cayley snark, by Lemma 2.8, or the normal subgroup contains a, which is impossible by Lemma 3.3, or it has index 2). The reader should consult [33] for the original proof; the following short argument, based on the lemmas above and on Theorem 5.1, recovers the statement.

Proof of Theorem 6.1. Let $X = X(G; a, x)$ be a Cayley snark of smallest order; by Proposition 3.1, $s = \mathrm{ord}(x)$ is odd and G is not abelian. Let $1 \neq N \trianglelefteq G$ be a proper normal subgroup.

- If $a \notin N$ and $x \notin N$, then by Lemma 2.8 X covers the smaller cubic Cayley graph $X(G/N; \bar{a}, \bar{x})$, which is 3-edge-colourable by minimality; hence so is X, a contradiction.
- If $a \in N$, then X is 3-edge-colourable by Lemma 3.3, a contradiction.
- Hence $x \in N$ and $a \notin N$. Then $G/N = \langle aN \rangle$ has order 2.

So every non-trivial proper normal subgroup of G has index 2 and contains x. If $N_1 \neq N_2$ were two of them, $N_1 \cap N_2$ would be a normal subgroup of index 4 containing $x \neq 1$, contradicting what we just proved. Hence G has at most one non-trivial proper normal subgroup.

If it has none, G is simple, and non-abelian because $|G|$ is even and divisible by $s \geq 3$.

Otherwise let T be the unique non-trivial proper normal subgroup. It is a minimal normal subgroup, hence characteristically simple: $T \cong S^k$ for a simple group S. If S is abelian then $G = T \rtimes \langle a \rangle$ is solvable, contradicting Theorem 5.1. So S is non-abelian simple, and the k direct factors of T are the minimal normal subgroups of T, which a permutes by conjugation. The product of the factors in one $\langle a \rangle$-orbit is normalised by T and by a, hence normal in G, hence equal to T; so a acts transitively on the k factors and $k \leq 2$. If $k = 1$, T is non-abelian simple and G is almost simple. If $k = 2$, a interchanges the two factors; after replacing the second factor's coordinates by their image under a suitable automorphism of S we may assume that a acts by $(u, v) \mapsto (v, u)$, i.e. $G \cong S \wr \mathbb{Z}_2$. □

Remark 6.2. (1) In case (ii) the graph X is the bi-Cayley graph $B(T; x, \alpha(x))$ of (1), where α is the outer involutory automorphism of T induced by a (it is not inner: if a induced an inner automorphism ι_t, then at^{-1} would be a central involution of G, generating a second normal subgroup). Examples of admissible pairs (G, T): (S_n, A_n) with a an odd permutation; $(\mathrm{PGL}(2, q), \mathrm{PSL}(2, q))$ for odd q; $(\mathrm{P\Sigma L}(2, q^2), \mathrm{PSL}(2, q^2))$; (M_{10}, A_6); $(S \wr \mathbb{Z}_2, S \times S)$ with, for instance, $S = A_5$ and $x = ((123), (12345))$.

(2) Groups of type (i) or (ii) are precisely the groups $G = \langle a, x \rangle$ whose *socle* is a non-abelian minimal normal subgroup T with G/T of order at most 2; this explains the title of [33]. Note that the theorem does not say that every Cayley snark lives on such a group, only that a *smallest* one does; a Cayley snark on an arbitrary group covers one of these.

(3) The argument shows more generally that Conjecture 1.1 holds for every group G that has a non-trivial normal subgroup N with $a \in N$ or with $a, x \notin N$ and the conjecture known for G/N. For instance, it holds for every group of the form $G = N \rtimes \mathbb{Z}_m$ with m odd and N containing a, and for all groups G with a solvable non-trivial normal subgroup N such that G/N satisfies the conjecture (by induction, if $a, x \notin N$; by Lemma 3.3 if $a \in N$; and if $x \in N$ then $|G : N| = 2$ and G is solvable). In particular, the conjecture holds for all groups with a non-trivial solvable radical whose quotient by the radical satisfies it.

7 The hamiltonicity route and Cayley maps

As noted in the introduction, a Hamilton cycle in a cubic Cayley graph yields a 3-edge-colouring. The literature on Hamilton cycles in Cayley graphs is vast (see the surveys [11, 24, 34]); we mention only the results that bear directly on cubic Cayley graphs on non-solvable groups, since for solvable groups Theorem 5.1 is already available.

7.1 $(2, s, 3)$-Cayley graphs

Glover and Marušič [15] initiated the study of cubic Cayley graphs $X(G; a, x)$ on groups with a $(2, s, 3)$-*presentation*

$$G = \langle a, x \mid a^2 = x^s = (ax)^3 = 1, \dots \rangle,$$

i.e. generated by an involution a and an element x of order s whose product has order 3. Such a graph has a natural embedding as a *Cayley map* in an orientable surface whose faces are the s-gons bounded by the x-cycles and the hexagons bounded by the closed walks $(ax)^3$; a Hamilton cycle is sought as the boundary of a tree of faces (a "tree of faces" in the Cayley map), using classical results on cyclic stability of cubic graphs.

Theorem 7.1 (Glover–Marušič [15]; Glover–Kutnar–Marušič [13]; Glover–Kutnar–Malnič—Marušič [14])**.** *Let $X = X(G; a, x)$ where G has a $(2, s, 3)$-presentation. Then X has a Hamilton path, and it has a Hamilton cycle whenever $|G| \equiv 2 \pmod 4$, whenever $s \equiv 0 \pmod 4$, and whenever s is odd. The only open case is $|G| \equiv 0 \pmod 4$ with $s \equiv 2 \pmod 4$.*

Corollary 7.2. *Every $(2, s, 3)$-Cayley graph is 3-edge-colourable.*

Proof. If s is even, Proposition 3.1(b) applies; if s is odd, Theorem 7.1 provides a Hamilton cycle. □

Note that a $(2, s, 3)$-generated group is the same thing as a $(2, 3)$-generated group (put $y = ax$), but the Cayley graph $X(G; a, x)$ of Corollary 7.2 is different from the graph $X(G; a, y)$ of §3.4: in the former the element of order 3 is the *product* of the two generators. The two families overlap exactly when $s = 3$.

7.2 Cayley maps and independent sets

Hujdurović, Kutnar and Marušič [16, 17] pursued the Cayley-map approach with Conjecture 1.1 rather than hamiltonicity as the target. In [16] they show that a connected bicirculant of prime valency admitting a suitable automorphism is *near-bipartite* (its vertex set has an independent subset whose removal leaves a bipartite graph), and combine this with the theory of Cayley maps to obtain partial results on the non-existence of Cayley snarks; [17] develops this further, "combining the theory of Cayley maps with the existence of certain kinds of independent sets of vertices in arc-transitive graphs" to obtain "new partial results suggesting promising future research directions" for the conjecture. The underlying idea is that, in $X(G; a, x)$ with s odd, the x-cycles are the vertices of an s-valent G-arc-transitive quotient graph (the coset graph of $\langle x \rangle$), and a suitable independent set in this quotient allows one to modify the odd 2-factor F_x into an even one.

7.3 Other hamiltonicity results

Kutnar, Marušič, Morris, Morris and Šparl [22] proved that every connected Cayley graph whose order is kp with $k < 32$, $k \neq 24$ (p prime), or kpq with $k \leq 5$, or pqr, or kp^2 with $k \leq 4$, or kp^3 with $k \leq 2$ (p, q, r distinct primes) is hamiltonian, and Morris and Wilk [23] extended the first case to all $k < 48$; in particular every connected Cayley graph of order less than 48 is hamiltonian, and so are, for instance, the cubic Cayley graphs of order 60 on A_5 and of order 120 on S_5 and $\mathrm{SL}(2, 5)$. All of these are, however, well inside the range of the census discussed next.

A different kind of result concerns a single, very natural family: for $\sigma = (1\,2 \cdots n)$ and $\tau = (1\,2)$ the graph $X(S_n; \tau, \sigma)$ is the undirected "sigma-tau graph". Compton and Williamson [6] showed that it is hamiltonian for every $n \geq 3$, and Sawada and Williams [39] solved the much harder directed version (Nijenhuis–Wilf's sigma-tau problem): the Cayley *digraph* on $\{\sigma, \tau\}$ has a Hamilton cycle if and only if n is odd. For odd n this is a bi-Cayley graph of the form (1) on $T = A_n$ with α conjugation by a transposition, so Conjecture 1.1 holds for these graphs; the case of a general odd-order $x \in A_n$ is open.

8 Computational evidence

8.1 The census of cubic vertex-transitive graphs

Potočnik, Spiga and Verret [38] constructed all connected cubic vertex-transitive graphs on at most 1280 vertices: there are 111 360 of them, of which 109 926 are Cayley graphs (97 687 of these are graphical regular representations, i.e. their full automorphism group acts regularly). They report:

> "We tested all the graphs in the census for Hamilton cycles and found at least one such cycle for each graph apart from the four well-known exceptions (the Petersen graph, the Coxeter graph and their truncations)." [38, §2]

Corollary 8.1. *Every connected cubic Cayley graph on at most* 1280 *vertices is* 3*-edge-colourable. More generally, every connected cubic vertex-transitive graph on at most* 1280 *vertices other than the Petersen graph and its truncation is* 3*-edge-colourable.*

Proof. A hamiltonian cubic graph of even order is 3-edge-colourable. The Coxeter graph has chromatic index 3, hence so does its truncation by Lemma 3.5; the Petersen graph and its truncation are not 3-edge-colourable, by Lemma 3.5 again. None of the four exceptions is a Cayley graph. □

8.2 Beyond the census

Combining Corollary 8.1 with Theorem 6.1 gives a concrete list of the smallest groups on which a Cayley snark could conceivably live: a smallest Cayley snark has more than 1280 vertices, and its group is a non-abelian simple group of order > 1280, or an almost simple group $T.2$ with $|T| > 640$ and T simple, or $S \wr \mathbb{Z}_2$ with $|S| > \sqrt{320}$, i.e. $|S| \geq 60$. In increasing order the first candidates are $\mathrm{PGL}(2, 11)$ (order 1320), $\mathrm{PGL}(2, 13)$ (2184), $\mathrm{PSL}(2, 17)$ (2448), A_7 (2520), $\mathrm{PSL}(2, 19)$ (3420), $\mathrm{PSL}(2, 16)$ (4080), $\mathrm{PGL}(2, 17)$ (4896), S_7 (5040), $\mathrm{PSL}(3, 3)$ (5616), $\mathrm{PSU}(3, 3)$ (6048), $\mathrm{PSL}(2, 23)$ (6072), $\mathrm{PGL}(2, 19)$ (6840), $A_5 \wr \mathbb{Z}_2$ (7200), $\mathrm{PSL}(2, 25)$ (7800), M_{11} (7920), $\mathrm{P\Sigma L}(2, 16)$ (8160), . . .

We wrote a program (`code/cayley_snark_check2.py` in the repository accompanying this survey; an earlier version, `cayley_snark_check.py`, is described below) which, for a given permutation group G, enumerates all pairs (a, x) with a an involution, x of odd order and $\langle a, x\rangle = G$, up to conjugation in G and inversion of x, and decides whether $X(G; a, x)$ is 3-edge-colourable. By Proposition 3.1 this decides Conjecture 1.1 for *all* cubic Cayley graphs on G. The decision procedure is Lemma 4.1: for a subgroup $H \leq G$ containing no conjugate of x, the quotient pregraph $H\backslash X$ has $|G : H|$ vertices, and a 3-edge-colouring of it—found by the SAT solver CaDiCaL [7] through the python-sat interface, with one Boolean variable "e has colour 2" and one variable "e has colour 1" per edge or semi-edge e and the obvious clauses at each vertex—lifts to an H-invariant 3-edge-colouring of X. The lifted colouring is then verified independently on X: for every $g \in G$ the program checks that the three edges at g have three different colours and that the colour of $\{g, ga\}$ is the same when computed from g and from ga. Candidate subgroups H are generated once per group—point stabilisers, pointwise and setwise stabilisers of pairs and of small sets, stabilisers of partitions into blocks of equal size, centralisers and normalisers of elements of odd order and of involutions, and abelian subgroups of odd order—(conjugate subgroups give isomorphic quotients, so at most two candidates with the same order and the same multiset of element orders are kept), and for each pair (a, x) those not meeting the conjugacy class of x are tried in decreasing order of size, the full Cayley graph (with a time limit) serving as a last resort; the last resort was never needed. Generation is tested through the socle T of G ($T = G$ or $|G : T| = 2$): $\langle a, x\rangle = G$ if and only if $a \notin T$ and $\langle x, x^a\rangle = T$ (or $\langle a, x\rangle = T$), and a subgroup of T generated by two given elements is all of T as soon as its order exceeds $|T|/m$, where m is the minimal index of a proper subgroup of T (its minimal degree, taken from the Atlas). Groups are given by permutation generators (constructed from matrices over finite fields, or copied from the Atlas of Finite Group Representations [8]), and the program checks every group order against the known value; the Petersen graph is rejected as a control, and the SAT encoding for pregraphs was cross-checked against exhaustive search on a thousand random small pregraphs.

The effect of Lemma 4.1 is dramatic. The earlier program, which only used abelian subgroups H of odd order (no semi-edges), reduced $X(S_8; a, x)$ with x of order 15 to graphs on 4480 vertices on which the solver timed out, leaving 19 pairs on S_8 undecided; with $H = S_4 \wr S_2$ (the stabiliser of a partition into two blocks of four, which contains no element of cycle type $5 \cdot 3$) the quotient has 35 vertices and is coloured instantly. The largest quotient that had to be solved for any group in Table 2 has 9 240 vertices, whereas the groups have order up to 259 200. Table 2 lists, for each group, the number of generating pairs, the orders of x that occur, and the number $\max |\bar{V}|$ of vertices of the largest quotient pregraph that was needed; all 18 056 Cayley graphs were found to be 3-edge-colourable.

Theorem 8.2. *Conjecture 1.1 holds for every cubic Cayley graph on each of the 81 groups listed in Table 2; in particular for every non-abelian simple group of order at most* 246 480*, for* $\mathrm{PSL}(2, q)$ *with* $q \leq 61$ *and* $q \in \{67, 71, 73, 79\}$*, for* $\mathrm{PGL}(2, q)$ *with* $q \leq 61$*, for* A_n *with* $n \leq 9$ *and* S_n *with* $n \leq 8$*, and for the groups* M_{10}, $\mathrm{PSL}(3, 3)$, $\mathrm{PSU}(3, 3)$, $A_5 \wr \mathbb{Z}_2$, M_{11}, $\mathrm{P\Sigma L}(2, 16)$, $\mathrm{PSL}(3, 3).2$, $G_2(2) = \mathrm{PSU}(3, 3).2$, $\mathrm{P\Sigma L}(2, 25)$, $\mathrm{PSL}(2, 25).2_3$, $\mathrm{PSL}(3, 4)$, $\mathrm{PSU}(4, 2) = \mathrm{PSp}(4, 3)$, $Sz(8)$, $\mathrm{PSL}(3, 4).2$ *(field)*, $\mathrm{PSL}(3, 4).2$ *(graph)*, $\mathrm{PSL}(3, 4).2$ *(graph-field)*, $\mathrm{PSU}(4, 2).2 = W(E_6)$, $\mathrm{PSL}(2, 7) \wr \mathbb{Z}_2$, $\mathrm{PSU}(3, 4)$, M_{12}, $\mathrm{PSL}(2, 49).2_3$, $\mathrm{P\Sigma L}(2, 49)$, $\mathrm{PSU}(3, 4).2$, $\mathrm{PSU}(3, 5)$, J_1, $M_{12}.2$, $\mathrm{PSU}(3, 5).2$, $A_6 \wr \mathbb{Z}_2$.

Two of the groups in the table, M_{10} and $\mathrm{PSL}(2, 25).2_3$, have all their involutions inside the socle and are therefore not generated by an involution and an element of odd order at all; the only cubic Cayley graphs on them are the trivially colourable ones of Proposition 3.1. The groups permitted by Theorem 6.1 for a smallest Cayley snark are the non-abelian simple groups, their extensions $T.2$ of index two and the wreath products $S \wr \mathbb{Z}_2$. Listing these by order (from the orders of the simple groups; note that $\mathrm{PSL}(2, 8)$, $Sz(8)$ and $\mathrm{PSL}(2, 32)$ have no extension of index two, while $\mathrm{PSL}(2, 25)$, $\mathrm{PSL}(2, 49)$ and $\mathrm{PSL}(3, 4)$ have three each), one finds that every such group of order less than 262 080 appears in Table 2, the first ones missing being $\mathrm{PSL}(2, 64)$ (order 262 080), $\mathrm{PSL}(2, 81)$ (order 265 680), $\mathrm{PSL}(2, 83)$ (order 285 852), $\mathrm{PGL}(2, 67)$ (order 300 696). Combining Theorem 6.1 with Corollary 8.1 and Table 2 therefore gives:

Corollary 8.3. *A Cayley snark of smallest order, if one exists, has more than* 262 079 *vertices. Its group is a non-abelian simple group, or a group* $T.2$ *or* $S \wr \mathbb{Z}_2$ *as in Theorem 6.1*(ii)*, of order at least* 262 080 *and not listed in Table 2.*

We stress that these computations, like the census, are evidence rather than insight: the even 2-factors found by the solver do not seem to follow any pattern that would suggest a general construction. One structural remark can be made, though. In every case a colouring invariant under a fairly large subgroup was found—the quotient never had more than 9 240 vertices—and for the groups $\mathrm{PSL}(2, q)$ with x in a non-split torus the $(q+1)$-vertex quotient by a Borel subgroup sufficed for the great majority of pairs once $q \geq 25$ (Example 4.3); but for small q, and for q even, it does not, so "the" natural quotient is not the answer. Another instructive family is $\mathrm{PGL}(2, q)$ with $\mathrm{ord}(x) = 3$ and $\mathrm{ord}(ax) = q + 1$: the quotients by the dihedral subgroups of order $2(q + 1)$ (admissible when $3 \nmid q + 1$) were never 3-edge-colourable—for $q = 61$ the solver proved all 30 such quotients, on 1830 and 3660 vertices, uncolourable—whereas the quotient by the unipotent subgroup of order q, or by a Borel subgroup when $3 \nmid q - 1$, always was.

9 Related problems on flows in Cayley graphs

The flow formulation of Conjecture 1.1 has stimulated work on the stronger 3-flow property. By Mader's theorem (Theorem 2.4), Tutte's 3-flow conjecture restricted to vertex-transitive graphs asserts that every vertex-transitive graph of valency at least 4 admits a nowhere-zero 3-flow. This is known for Cayley graphs on abelian groups (Potočnik, Škoviera and Škrekovski [37]), for Cayley graphs of valency at least 4 on groups with a certain Sylow 2-subgroup structure (Nánásiová and Škoviera [30]), for graphs admitting a solvable arc-transitive group of automorphisms (Li and Zhou [25]), for Cayley graphs on supersolvable groups (Zhang and Zhou [45]), for nilpotently vertex-transitive graphs [44], for various dihedral and generalised dihedral and quaternion groups [43, 2], and for Cayley graphs on solvable groups of twice square-free order [3]. The general case for solvable groups, which would be the 3-flow analogue of Theorem 5.1, is open. These results do not bear on the cubic case (a cubic graph never has a nowhere-zero 3-flow), but their proofs use the same quotient-and-lift technique of §2.5.

10 Open problems

1. (Conjecture 1.1 for simple groups.) Is $X(G; a, x)$ 3-edge-colourable for every non-abelian simple group $G = \langle a, x\rangle$, $a^2 = 1$, $\mathrm{ord}(x)$ odd? Even the case $G = \mathrm{PSL}(2, q)$ or $G = A_n$ for all q, n is open, although Corollary 7.2 covers those generating pairs with $\mathrm{ord}(ax) = 3$.

2. (The bi-Cayley case.) For a non-abelian simple group T with an outer involutory automorphism α and $x \in T$ of odd order with $\langle x, \alpha(x)\rangle = T$, is $B(T; x, \alpha(x))$ always 3-edge-colourable? The case $T = A_n$, α conjugation by a transposition, is already interesting. Same question for $T = S \times S$ with α the swap.

3. (Arc-regular cubic graphs.) Is every connected cubic graph admitting an arc-regular group of automorphisms 3-edge-colourable (Corollary 3.7)? By Corollary 3.8 the answer is yes up to 10 000 vertices.

4. (Vertex-transitive graphs.) Are the Petersen graph and its truncation the only connected cubic vertex-transitive graphs that are not 3-edge-colourable? Corollary 8.1 says yes up to 1280 vertices, and Theorem 5.2 says yes for solvable vertex-transitive groups.

5. (Girth.) Does Conjecture 1.1 hold for cubic Cayley graphs of girth 3 and 5 (equivalently, by Proposition 3.4, for generators x of order 3 and 5), and of girth 6? Does it hold for girth at least 7, i.e. is there a cyclically 7-edge-connected Cayley snark?

6. (Hamiltonicity.) Is every $(2, s, 3)$-Cayley graph with $|G| \equiv 0 \pmod 4$ and $s \equiv 2 \pmod 4$ hamiltonian? (This is the remaining case of Theorem 7.1; it does not affect Conjecture 1.1.)

7. (Structure.) Is there an algebraic characterisation of the even 2-factors of $X(G; a, x)$, for instance in terms of a subgroup H of odd order such that the colouring is H-invariant? Our computations show that in all cases tried an H-invariant colouring exists for a subgroup H of index at most 9 240 (Table 2), often a point stabiliser or an involution centraliser; is there always an H-invariant colouring with H of bounded index (say $|G : H| \leq c\,|G|^{1/2}$), or with H from a specific family such as the centraliser of an involution?

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group G	$\lvert G\rvert$	inv. classes	pairs (a,x)	orders of x	max $\lvert\bar{V}\rvert$	result
A_5	60	1	6	3, 5	15	all colourable
S_5	120	2	2	5	5	all colourable
$\mathrm{PSL}(2,7)$	168	1	4	3, 7	21	all colourable
$\mathrm{PGL}(2,7)$	336	2	5	3, 7	42	all colourable
A_6	360	1	4	5	10	all colourable
$\mathrm{PSL}(2,8)$	504	1	42	3, 7, 9	63	all colourable
$\mathrm{PSL}(2,11)$	660	1	14	3, 5, 11	66	all colourable
M_{10}	720	1	0	–	–	no such pairs
$\mathrm{PGL}(2,9)$	720	2	10	3, 5	45	all colourable
S_6	720	3	2	5	10	all colourable
$\mathrm{PSL}(2,13)$	1092	1	45	3, 7, 13	84	all colourable
$\mathrm{PGL}(2,11)$	1320	2	17	3, 5, 11	120	all colourable
$\mathrm{PGL}(2,13)$	2184	2	26	3, 7, 13	168	all colourable
$\mathrm{PSL}(2,17)$	2448	1	60	3, 9, 17	153	all colourable
A_7	2520	1	11	5, 7	35	all colourable
$\mathrm{PSL}(2,19)$	3420	1	76	3, 5, 9, 19	180	all colourable
$\mathrm{PSL}(2,16)$	4080	1	204	3, 5, 15, 17	255	all colourable
$\mathrm{PGL}(2,17)$	4896	2	36	3, 9, 17	288	all colourable
S_7	5040	3	19	3, 5, 7	42	all colourable
$\mathrm{PSL}(3,3)$	5616	1	20	3, 13	78	all colourable
$\mathrm{PSU}(3,3)$	6048	1	6	7	28	all colourable
$\mathrm{PSL}(2,23)$	6072	1	107	3, 11, 23	276	all colourable
$\mathrm{PGL}(2,19)$	6840	2	58	3, 5, 9, 19	360	all colourable
$A_5 \wr \mathbb{Z}_2$	7200	3	8	15	25	all colourable
$\mathrm{PSL}(2,25)$	7800	1	134	3, 5, 13	300	all colourable
M_{11}	7920	1	14	5, 11	55	all colourable
$\mathrm{P\Sigma L}(2,16)$	8160	2	20	15, 17	68	all colourable
$\mathrm{PSL}(2,27)$	9828	1	186	3, 7, 13	378	all colourable
$\mathrm{PSL}(3,3).2$	11232	2	31	3, 13	156	all colourable
$G_2(2) = \mathrm{PSU}(3,3).2$	12096	2	21	3, 7	378	all colourable
$\mathrm{PGL}(2,23)$	12144	2	72	3, 11, 23	528	all colourable
$\mathrm{PSL}(2,29)$	12180	1	219	3, 5, 7, 15, 29	435	all colourable
$\mathrm{PSL}(2,31)$	14880	1	160	3, 5, 15, 31	480	all colourable
$\mathrm{PGL}(2,25)$	15600	2	90	3, 5, 13	300	all colourable
$\mathrm{PSL}(2,25).2_3$	15600	1	0	–	–	no such pairs
$\mathrm{P\Sigma L}(2,25)$	15600	3	30	13	26	all colourable
$\mathrm{PGL}(2,27)$	19656	2	123	3, 7, 13	378	all colourable
A_8	20160	2	33	5, 7, 15	35	all colourable
$\mathrm{PSL}(3,4)$	20160	1	60	5, 7	105	all colourable
$\mathrm{PGL}(2,29)$	24360	2	146	3, 5, 7, 15, 29	840	all colourable
$\mathrm{PSL}(2,37)$	25308	1	378	3, 9, 19, 37	684	all colourable
$\mathrm{PSU}(4,2) = \mathrm{PSp}(4,3)$	25920	2	42	5, 9	120	all colourable
$Sz(8)$	29120	1	210	5, 7, 13	455	all colourable
$\mathrm{PGL}(2,31)$	29760	2	113	3, 5, 15, 31	960	all colourable
$\mathrm{PSL}(2,32)$	32736	1	930	3, 11, 31, 33	1023	all colourable
$\mathrm{PSL}(2,41)$	34440	1	380	3, 5, 7, 21, 41	861	all colourable
$\mathrm{PSL}(2,43)$	39732	1	490	3, 7, 11, 21, 43	924	all colourable
$\mathrm{PSL}(3,4).2$ (field)	40320	2	11	5, 7	105	all colourable
$\mathrm{PSL}(3,4).2$ (graph)	40320	2	68	3, 5, 7	315	all colourable
$\mathrm{PSL}(3,4).2$ (graph-field)	40320	2	45	5, 7	105	all colourable
S_8	40320	4	47	5, 7, 15	35	all colourable
$\mathrm{PGL}(2,37)$	50616	2	242	3, 9, 19, 37	1368	all colourable
$\mathrm{PSU}(4,2).2 = W(E_6)$	51840	4	52	5, 9	40	all colourable
$\mathrm{PSL}(2,47)$	51888	1	437	3, 23, 47	1128	all colourable
$\mathrm{PSL}(2,7) \wr \mathbb{Z}_2$	56448	3	6	21	42	all colourable
$\mathrm{PSL}(2,49)$	58800	1	480	3, 5, 7, 25	1200	all colourable
$\mathrm{PSU}(3,4)$	62400	1	64	3, 5, 13, 15	195	all colourable
$\mathrm{PGL}(2,41)$	68880	2	250	3, 5, 7, 21, 41	1680	all colourable
$\mathrm{PSL}(2,53)$	74412	1	780	3, 9, 13, 27, 53	1431	all colourable
$\mathrm{PGL}(2,43)$	79464	2	325	3, 7, 11, 21, 43	1848	all colourable
M_{12}	95040	2	66	3, 5, 11	396	all colourable
$\mathrm{PSL}(2,59)$	102660	1	926	3, 5, 15, 29, 59	1770	all colourable
$\mathrm{PGL}(2,47)$	103776	2	288	3, 23, 47	2208	all colourable